SPE-227150-MS

Advancing Gas Rate Estimation: Integrating Physics into Machine Learning Models

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Accurate gas flow rate estimation plays a pivotal role in optimizing well performance and field operations. Traditional empirical models, such as the Gilbert equations and its modified forms, often fall short in capturing the complexity of multiphase flow and compositional effects under varying operational conditions. Pure data-driven machine learning (ML) models offer high accuracy but suffer from a lack of physical interpretability and poor extrapolation in edge cases. This study presents a Physics-Informed Machine Learning (PIML) framework that integrates a recalibrated Modified Gilbert equation into a hybrid ensemble ML model to enhance both predictive accuracy and physical consistency.

Using a field dataset of 1,181 test records, the approach employs feature engineering, hybrid loss functions, and ensemble stacking (XGBoost, MLP, Ridge) to incorporate domain physics into the learning loop. The recalibrated choke equation is embedded into model training via a custom loss function to anchor predictions around physical expectations. The model achieves an R^2 of 0.973 and an MSE of 0.393—matching the accuracy of pure ML models while gaining robustness in sparse data regions. Uncertainty quantification with bootstrapping and quantile regression validates prediction confidence, while SHAP analysis and feature importance plots offer transparency into the model's decision-making process. The results demonstrate that physics-informed models can bridge the gap between analytical heuristics and black-box ML by offering scalable, accurate, and explainable gas rate estimation solutions. This work lays the foundation for intelligent virtual flow metering systems applicable in real-time production monitoring and optimization.

Introduction and Background

Conventional gas rate estimation in well operations has heavily relied on analytical equations such as the Rawlins and Schellhardt (1935), Gilbert equation (1954) and its modified forms which related gas rate to tubing head pressure and choke size assuming critical flow. Subsequent variants incorporated temperature, specific gravity, and Condensate Gas Ratio (CGR) effects (Hagedorn and Brown 1965; Leal et al. 2013). While these empirical correlations are simple to apply and widely adopted in field operations, they lack adaptability across changing reservoir and surface conditions. Specifically, choke-based correlations often suffer under multiphase flow regimes, variable CGR environments, and high water cut scenarios, leading to suboptimal predictions and potential operational inefficiencies.

Building on the work presented in BinZiad et al. (2024), where data-driven machine learning (ML) models outperformed choke equations in gas flow prediction accuracy, the limitations of purely empirical models were further highlighted. However, the prior study also indicated that pure ML models, although accurate, lacked interpretability and occasionally produced unrealistic forecasts under edge conditions. This motivates the integration of domain knowledge through Physics-Informed Machine Learning (PIML), where governing physical equations serve as an upstream guide to the learning process. The current work advances this effort by embedding a recalibrated Modified Gilbert equation into the ML training process via a custom loss function. The objective is to enhance prediction accuracy while preserving physical realism and model explainability.

Literature Review

A growing trend involves embedding domain constraints into ML workflows, either through loss functions (PIML), architecture (PINNs), or feature engineering. PIML embeds known physical laws into machine learning models to improve interpretability and generalization. Techniques include neural architectures with embedded differential equations (Latrach et al. 2023). Unlike PINNs that require full PDE formulations, the PIML strategy adopted here is modular and retrofittable, allowing classical flow models to serve as soft constraints during model training. This architecture makes the solution more compatible with existing production environments, particularly in reservoirs where multiphase interactions are dominant but field-wide physics may be generalized using empirical surrogates.

A prominent implementation is hybrid Virtual Flow Metering (VFM), where firmware-based models are driven by multiphase flow physics and further refined by machine learning algorithms. VFMs are widely used in the oil and gas industry to estimate production rates without the need for intrusive hardware. These tools are typically classified into three categories: mechanistic models, data-driven models, and hybrid approaches (Amin, 2015). Mechanistic VFMs rely on detailed representations of multiphase flow physics, incorporating pressure-volume-temperature (PVT) data and wellbore hydraulics. However, their accuracy can be significantly compromised in the presence of challenging flow conditions such as sand production, scaling, or wax deposition, which are difficult to model explicitly. On the other hand, machine learning (ML)-based VFMs provide a flexible and often highly accurate alternative, but they require large volumes of labeled training data and may exhibit limited extrapolative capability when applied to unseen operating regimes. To overcome these limitations, hybrid VFMs have been proposed, which integrate simplified physical models with online learning, adaptive filtering, or Bayesian neural network techniques to improve robustness and adaptability (Grimstad et al. 2021). Nevertheless, the optimal method for coupling data-driven learning with physical constraints remains an area of active research, with varying levels of complexity and implementation maturity observed across current literature. Another example is the work by Henriksson et al. (2022), which demonstrated that combining physics-based VFM with ML led to significant gains in real-time multiphase flow accuracy.

Physics-informed neural networks (PINNs) have also demonstrated robustness in reservoir simulation tasks, outperforming purely data-driven models through physics-constrained loss terms. Nagao et al. (2024) developed a PINN-based framework for reservoir connectivity identification and production forecasting, showing notable improvements over standard ML and reduced-physics models. In the domain of hydraulic fracturing, a four part paper series comprehensively captures the critical role played by physics based datasets integrated with real-field data through transfer learning for enhancing model accuracy and generalization capabilities (Khan et al. 2024a; 2024b; 2025a; 2025b). Similarly, Bozoev et al. (2025) propose a surrogate modeling approach for production prediction purely based on physical reservoir simulations.

Methodology

A novel approach was employed with the structured integration of a recalibrated Modified Gilbert equation as a physics-based prior into a stacked ML pipeline (XGBoost + MLP + Ridge), with explainability verified through SHAP analysis. This dual emphasis on performance and interpretability aligns with industry priorities and addresses known limitations in generalized ML approaches for petroleum flow modeling. Fig. 1 presents the overall workflow of the physics-informed machine learning (PIML) methodology for gas rate estimation. It begins with SCADA or separator test data ingestion, followed by preprocessing, physics-based feature engineering, and empirical model recalibration. The workflow integrates hybrid loss-based ML training, ensemble stacking, and uncertainty quantification before deployment through a lightweight API interface for real-time inference.

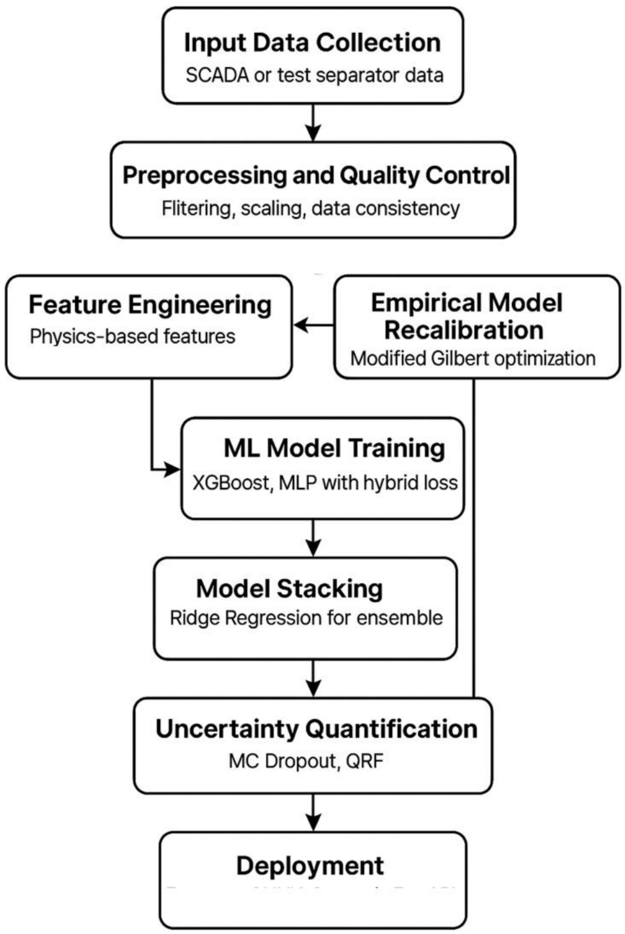

Figure 1—ML Modeling methodology from input data collection to deployment for production.

Feature Engineering

This study utilized a thoroughly cleaned and standardized dataset (prepared in the previous work by the authors – Binziad et al. 2024), composed of measured gas well test records obtained from separator-based flowbacks. The dataset comprising of 1,181 cleaned observations incorporated a diverse set of operational variables that are commonly available in production testing environments. Core features included choke size expressed in 64ths of an inch (D_{64}), flowing wellhead pressure (FWHP), and rates of oil (q_o), water (q_w), and gas q_g production. In addition, compositional parameters such as the concentrations of hydrogen sulfide (H_2S) and carbon dioxide (CO_2), as well as the condensate-to-gas ratio (CGR), were included to account for fluid property influences on flow behavior. Hence, most of the data handling and wrangling was borrowed from the previous work.

To enhance the learning capability of the models, several domain-informed features were engineered. A key derived parameter, referred to as the "Choke Flow Estimate," was formulated as the product of choke size and the square root of FWHP. This formulation was inspired by the empirical structure of classical flow equations, which relates pressure and flow through nonlinear relationships. Additional features included interaction terms that captured the influence of CGR and the water-gas ratio (WGR), as well as normalized ratios between choke size and FWHP, enabling the models to better capture flow regime transitions and condensate banking effects.

Physics-Based Model Baseline

As a foundational step, several physics-based equations traditionally used in estimating gas flow were implemented to serve as baseline models. These included the original Gilbert equation (equation1 below), its modified form incorporating CGR dependence (equation2 below), and an empirical formulation denoted as "New regional equation" by Leal et al. (2013) in equation3 below. Each of these equations were evaluated using both default and recalibrated parameter sets.

$$P_{up} = C_1 \frac{q \times GOR^{C_2}}{D_{64}^{C_3}} \tag{1}$$

$$q_g = \frac{P_{up} D_{64}^{b_1}}{b_2 CGR^{b_3}} \tag{2}$$

$$q_g = \beta_1 D_{64}^{\beta_2} \left(\frac{P_{up}}{14.7}\right) \sqrt{\left(\frac{1}{\gamma_g T}\right) \beta_3 \left[\left(\frac{P_{down}}{P_{up}}\right)^{\beta_4} - \left(\frac{P_{down}}{P_{up}}\right)^{\beta_5}\right]} \tag{3}$$

Where, P_{up} is upstream pressure, P_{down} is downstream pressure, γ_g is the gas specific gravity, T is the upstream temperature, C_i, $b_{i,}$ are empirical coefficients.

As an example, for the modified Gilbert equation, the default process was tried and although historically this model has been used in gas flow estimation through surface chokes, the results did not show a high agreement with the actual data (Fig. 2, left). This is primarily because: (a) the empirical constants used in the equation are not generalized for the basin fluid models. The default parameters such as the exponents on pressure differential and choke size are based on limited or different regional datasets. Applying them blindly to a broader, more heterogeneous field dataset introduces large residuals. (b) Lack of consideration for fluid composition effects. The original equation does not incorporate gas gravity, temperature, condensate content, or other multiphase effects. These are critical in wells producing wet gas or condensate-rich fluids where pressure drops do not translate linearly into flow. (c) deviations in the nonlinear flow regime- for smaller choke sizes or high-pressure drop scenarios, the flow behavior becomes transitional or turbulent, leading to systematic faults in estimations. Hence the results show an MSE of ~11.9 and an R^2 of 0.116, failing to capture the full range of physics governing the dataset.

To enhance the model performance, a custom recalibration routine was implemented (Fig. 2, right). This subroutine performs bounded nonlinear least squares optimization to minimize the error between predicted and measured flow rates. Broadly, the technique is based on few steps – (a) Minimizing the objective function RMSE between the model-predicted gas rate and actual separator-tested gas rate. (b) Estimating parameters for coefficients (b_i) in the Modified Gilbert form. (c) Bounding the exponent ranges wherein optimization is constrained to physically meaningful bounds to prevent overfitting and ensure stable convergence.

Figure 2—Predicted versus actual gas rate using the modified Gilbert equation. (left) default equation; (right) recalibrated fit.

In summary, the process involves non-linear regression techniques, bounded optimization and logarithmic transformations, to align model outputs with observed test data. This recalibration adapts the original Gilbert formulation to the local dataset and allows for direct comparison between physically derived estimations and machine-learned predictions, while also identifying the parameter sensitivity of classical models. The recalibration managed to lower the MSE and increase the R^2.

Traditional ML models aim to minimize purely statistical loss metrics, such as mean squared error (MSE), without regard for physical consistency. While this often leads to high predictive accuracy on seen data, it can result in unphysical predictions under extrapolation or sparse sampling conditions. To enforce consistency with physical principles during training, a custom loss function was developed with a recalibrated form of the Modified Gilbert equation, a physically grounded model for choke-dominated gas flow. This function augmented the standard error metric with an additional penalty term derived from the difference between the machine learning model's gas rate predictions and those estimated using the recalibrated Modified Gilbert equation. This hybrid loss function served to penalize deviations from the physics-based estimate, effectively anchoring the learning process around known flow dynamics while allowing flexibility to accommodate data-driven corrections. By integrating physics into the learning objective, the model was guided away from unphysical extrapolations, particularly in data-sparse regions, resulting in improved robustness and interpretability. Equation 4 below gives the hybrid loss function for our study. The total loss is defined as a weighted combination of two components. The first term ensures that the model is trained to closely match the observed data, while the second term penalizes divergence from physical expectations. This dual-objective formulation anchors the learning process within a physically meaningful envelope and enhances model generalizability, especially under sparse or noisy data conditions. The physics-based term acts as a regularization constraint, discouraging overfitting to statistical noise.

$$L_{total} = \alpha \cdot MSE\left(y_{true}, y_{pred}\right) + (1-\alpha) \cdot MSE\left(y_{true}, y_{phys}\right) \tag{4}$$

Where, L_{total} is the total loss function, y_{true} is groud truth gas rate from separator data, y_{pred} is the predicted gas rate from the ML model, y_{phys} is the gas rate predicted by the recalibrated Gilbert Model, α is the weighting factor (set to 0.8 in this study based on cross-validation).

ML Modeling Strategy

To surpass the limitations of standalone models, an ensemble learning strategy was adopted. The architecture shown in Fig. 3 comprised two base learners, an XGBoost regressor configured with a physics-informed objective, and a multi-layer perceptron (MLP) with 3 hidden layers X 100 neurons each, designed to capture intricate patterns and nonlinear relationships among input variables. These base learners were integrated with a meta-learner, a Ridge regression model, that acted as the final aggregator. This linear regressor with

L2 regularization aggregates predictions from the base learners, helps reduce variance, and ensures smoother generalization by penalizing overly complex fits.

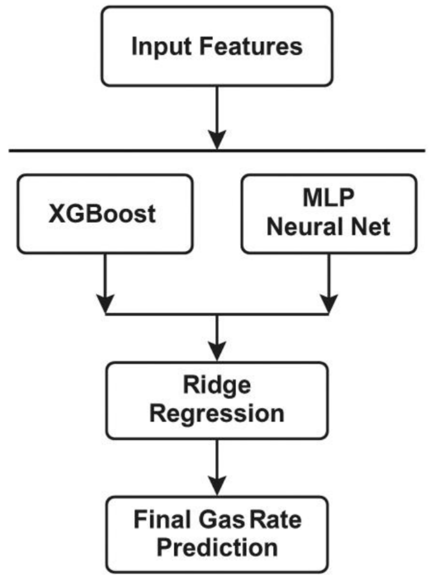

Figure 3—Ensemble ML model architecture with utilization of two base learners (trained in parallel using cross validation) and one meta learner.

Prior to training, input features were standardized using scaling transformations to ensure consistency in magnitude. Base models were trained in parallel using cross-validation. Predictions from these models were used as input features for the Ridge regressor. The entire pipeline was optimized using z-score normalization, polynomial interaction features, and custom hybrid loss function incorporating a physics-based penalty. Polynomial feature expansion up to the second degree was employed to capture interaction effects between variables. The model was evaluated using repeated cross-validation, and performance was quantified using established statistical metrics, MSE, MAE, and the R^2. This stacking method improved the predictive performance and interpretability of the model by combining the high-variance flexibility of MLP with the structured power of tree-based learners, while maintaining the benefits of linear aggregation. The loss function was implemented as a custom module within the MLP training loop using PyTorch and seamlessly integrated with the ensemble stacking architecture. The resulting model showed improved robustness, better residual distribution, and higher confidence in prediction intervals—particularly under unseen operational regimes.

Results

Fig. 4 shows the parity plot comparing predicted gas rates from the hybrid Physics-Informed Machine Learning (PIML) ensemble model against the actual separator-measured gas rates. The close clustering of points around the diagonal indicates excellent model accuracy and minimal bias across the entire operating range. Notably, the model maintains predictive fidelity at both low and high flow regimes, demonstrating strong generalization and consistency. This validates the effectiveness of the hybrid loss formulation and ensemble stacking strategy in capturing the complex non-linear relationships in multiphase gas production systems.

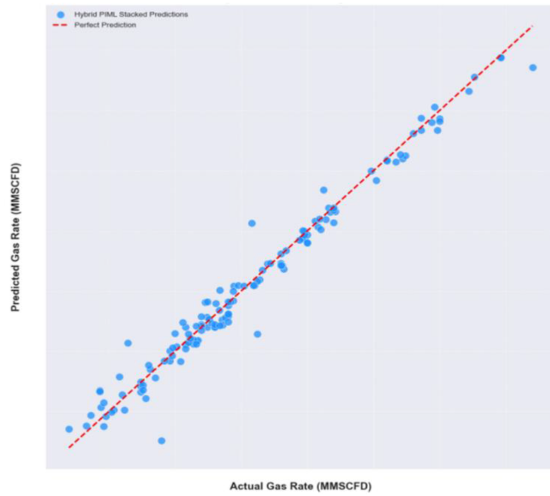

Figure 4—Actual versus predicted gas rate with a hybrid PIML stacking model.

Comparative Model Performance

A rigorous evaluation was conducted to compare the performance of various modeling strategies, ranging from traditional physics-based equations to standalone machine learning (ML) models and the proposed Physics-Informed Machine Learning (PIML) ensemble. The comparative results in Fig. 5 clearly indicate the superiority of the PIML stacking model, which integrates physics-consistent constraints into a layered ML framework. Specifically, the stacking model—comprising an XGBoost regressor with a physics-derived loss function and a multilayer perceptron, aggregated by a Ridge regression meta-learner—achieved an outstanding Mean Squared Error (MSE) of 0.393, a Mean Absolute Error (MAE) of 0.410, and an R^2 of 0.973. The performance metrics were slightly lower (almost on par) with a pure data-based XGBoost model (R^2 of approximately 0.985) but far superior compared to the pure physics model (low R^2). This suggests that tailored physics based models are very localized, non-scalable and can find significant issues converging in blind tests. Also, since the limitation of data-based models being overtrained and non-generalized is well-known, the inclusion of physics based loss functions into model training holds promise. Specifically, the XGBoost model lacked the embedded physics constraints that enable physical consistency and interpretability. This is particularly important in edge-case scenarios and sparse data regions. It is also important to note that the performance of XGBoost in this study pertains to a different data context compared to previous research (Binziad et al. 2024), which explains the slight variation in reported metrics.

Figure 5—Comparative performance evaluation metrics for baseline and hybrid models.

Model Evaluation

The residual distribution was analyzed to assess model generalization and potential bias. Residuals were plotted against the actual gas rates, as shown in Fig. 6. The plot revealed a symmetric and homoscedastic distribution, with no visible trend or systematic error across the range of flow rates. This pattern indicates that the model generalizes effectively across various flow conditions and is not overfitting to specific regimes or outliers. Such residual behavior is a desirable property of robust models, especially in production environments where operational variability can introduce significant uncertainty.

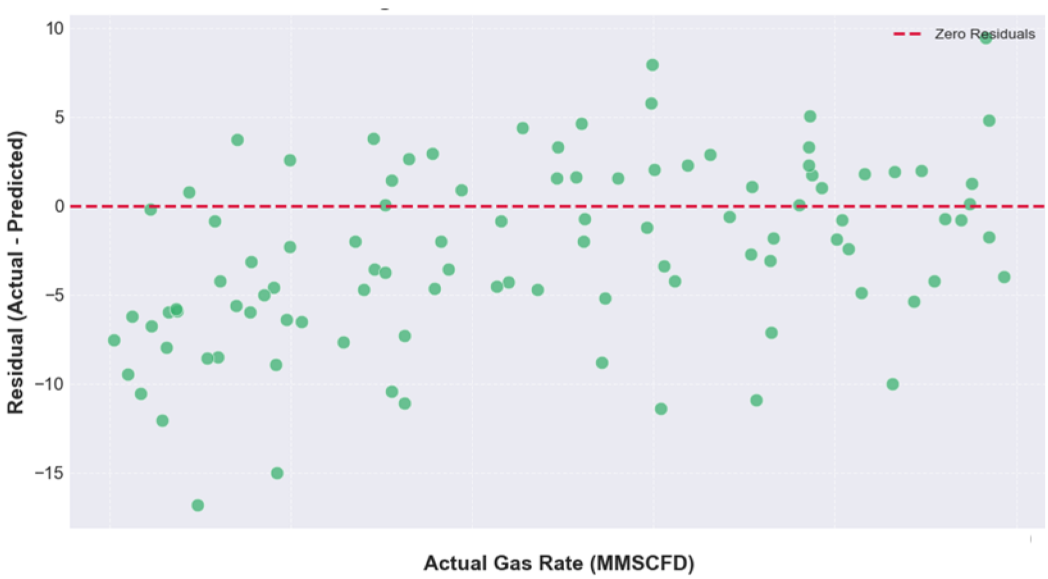

Figure 6—Residual plot for predicted versus actual gas rate; the residuals trend indicates generalization for PIML stacked model.

In real-world reservoir and production systems, the reliability of any machine learning (ML) model hinges not only on its point prediction accuracy but also on its ability to convey uncertainty in predictions. This is particularly critical in subsurface applications like gas rate forecasting, where sensor noise, dynamic flow conditions, and limited labeled data can introduce significant variability. Uncertainty quantification (UQ) thus becomes an essential component of model evaluation, offering insight into prediction confidence, and model robustness required for field deployment and operational decision-making.

To achieve this, we implemented two complementary techniques, bootstrapped residuals and quantile regression. The bootstrap method enables non-parametric estimation of prediction intervals by resampling model residuals, thereby capturing aleatoric uncertainty (arising from inherent noise or randomness in the data itself) without assuming a specific error distribution. On the other hand, quantile regression directly models conditional percentiles (e.g., 10th, 50th, 90th) of the target variable, providing a more asymmetric and data-adaptive view of uncertainty that naturally adjusts to varying noise levels across the input space.

Fig. 7 (top) illustrates the bootstrap-based 95% prediction interval, where the blue shaded region captures the spread of bootstrapped gas rate predictions around the unit slope reference line. The interval remains tight across most of the range, reflecting model consistency, with some widening in low and high gas rate regions which is likely due to sparser data. Fig. 7 (bottom) presents the 90% quantile-based prediction interval, showing a similarly aligned median line and an adaptive shaded region that adjusts to local prediction variance. Together, these methods enhance interpretability and provide a reliable envelope for model performance under uncertainty.

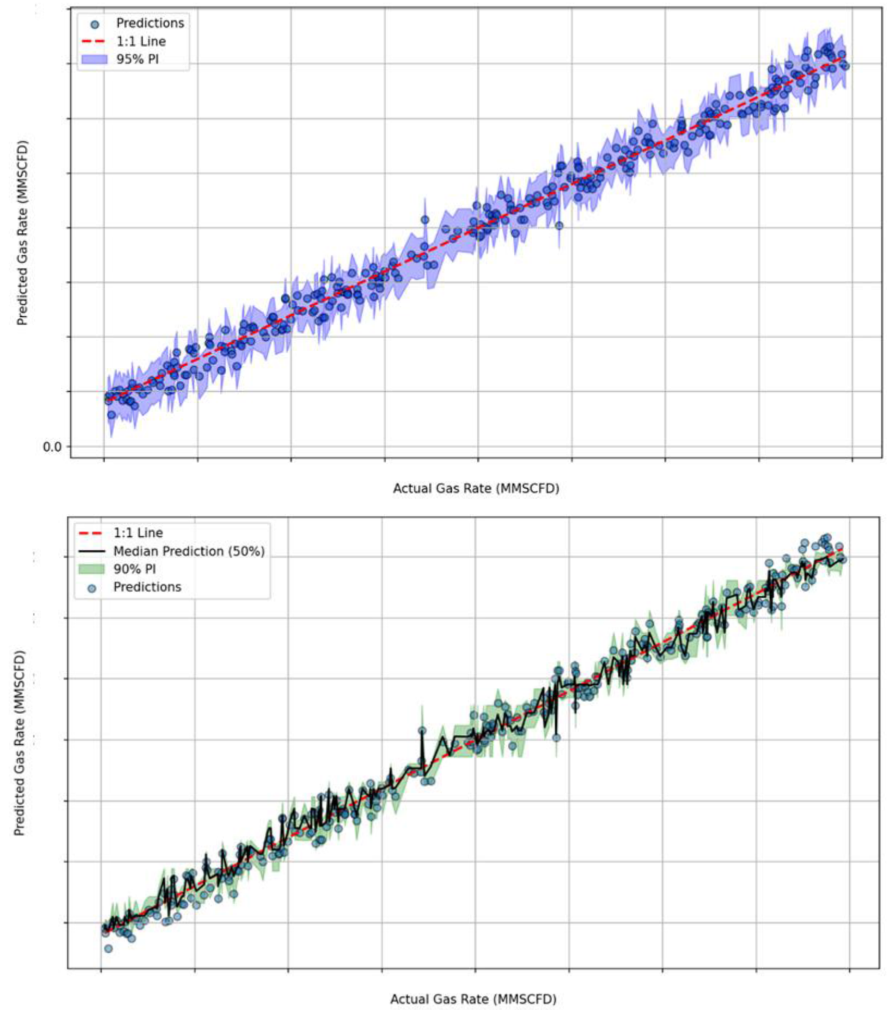

Figure 7—Uncertainty quantification for the PIML stacked model. (top) prediction interval via bootstrapped residuals; (bottom) prediction interval via quantile regression.

Model Explainability

Explainable AI is an important branch of ML modeling which hinges on the feature importance analysis. It is a critical component in interpreting models, especially in high-dimensional or physics-informed datasets. It helps domain experts understand which variables are driving the model predictions and enables improved model trust, refinement, and possible simplification.

The XGBoost feature importance plot quantifies feature relevance using the F-score (**Fig. X, top**), representing how frequently a feature is used in splits across all boosted trees. In this case, Oil condensate rate (BPD) and CGR (STB/MMSCFD) are overwhelmingly dominant, with F-scores of ~12.3 and ~3.6, respectively. These variables directly reflect liquid content and phase behavior, which are the dominant factors affecting the wellbore hydraulics, FWHP, and temperatures, making them intuitively vital for gas rate prediction. Other features such as H_2S, FWHP, and CO_2 show marginal contributions, possibly due to low variance or weak correlation with gas flow dynamics in this dataset. Interestingly, and counter-intuitively, choke size and choke flow estimate, often used in various empirical formulations, appear less influential in the ML formulation, further justifying the need for a data-driven or hybrid approach. The explanation for this feature ranking is that the top two features implicitly capture the others and if removed from training, the relevant features top the ranking.

To complement (and validate) this analysis, the SHAP (SHapley Additive exPlanations) calculations offers a more nuanced view, decomposing each model prediction to individual feature contributions. SHAP values, when visualized with a beeswarm plot (Fig. 8, bottom) also indicate directional impact—whether a high or low feature value increases or decreases gas rate predictions. For instance, high oil condensate and CGR values push predictions positively (rightward SHAP values), while extreme H_2S concentrations seem to slightly suppress them. The color gradient reinforces how feature value magnitude influences prediction direction. Together, these plots provide interpretability beyond pure black-box models and confirm that while traditional features may still play a role, physics-informed ML models derive most of their predictive power from fluid properties and condensate behavior rather than purely mechanical or empirical proxies.

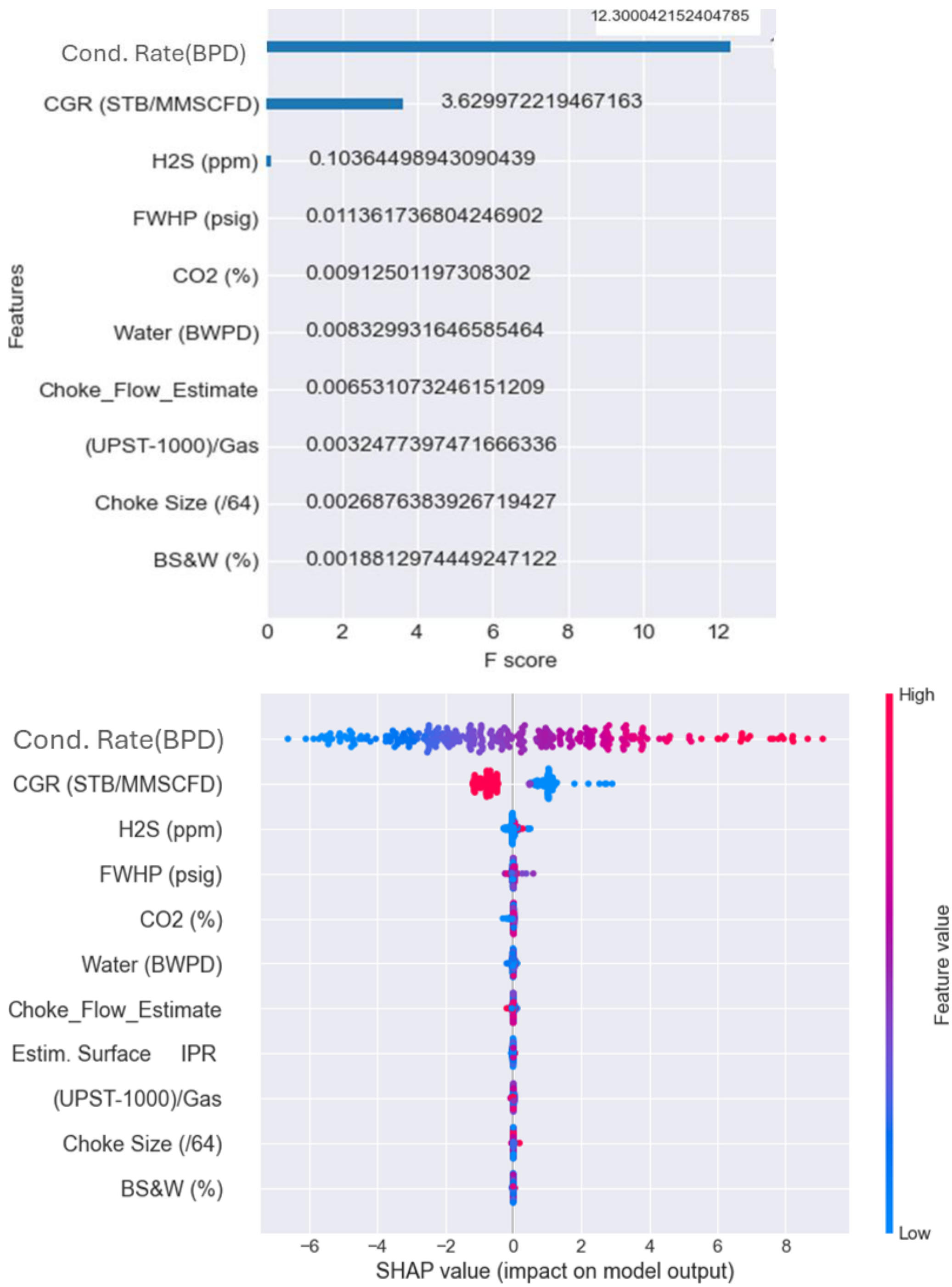

Figure 8—Feature importance evaluation for the PIML stacked model. (top) XGBoost standard loss with F-score; (bottom) beeswarm plot with Shapley Additive explanation (SHAP) values.

Discussion and Way Forward

While the proposed Physics-Informed Machine Learning (PIML) framework demonstrates strong performance for gas rate prediction, certain limitations and assumptions must be acknowledged to guide future improvements and real-world deployment. The recalibrated Modified Gilbert equation used as the physics backbone introduces a simplification by representing multiphase gas production through a single surrogate model. While this enhances interpretability and anchors the ML model in physical behavior, it may not capture the full complexity of flow regimes in systems with significant liquid loading, slugging, or condensate dropout. The assumption that the Gilbert-type equation is an adequate surrogate limits its applicability in more dynamic, non-steady systems or where flow regimes change rapidly. It needs to be expanded to a much larger library of physical equations in various flow domains. Secondly, the dataset itself, although rich in test points, is confined to a single asset with relatively consistent fluid properties and operating envelopes. This introduces a potential distributional bias and limits the generalizability of the model to other basins. No transfer learning, domain adaptation, or cross-field testing was performed in this study. As such, the robustness of the hybrid model across different geological or operational settings remains unverified. Also, The current study is focused on dry or wet gas environments. Adapting the same framework for oil-dominated or gas-condensate systems with severe slugging will require rethinking the physics surrogate or retraining loss formulation. After these applications, a broader evaluation across different asset types and basins is required to assess model scalability and transferability. Domain adaptation or transfer learning methods may help tailor the model for local conditions while preserving its physics-informed backbone.

The uncertainty quantification (UQ), while addressed through bootstrapping and quantile regression, was performed post-hoc and assumes that residual distributions are representative across all input conditions. This does not fully capture epistemic uncertainty, especially in data-scarce or out-of-distribution regions which would require Bayesian approaches or ensemble diversity methods to quantify. Nevertheless, the current UQ implementation adds practical value by offering confidence bands for predictions in the absence of more complex probabilistic modeling. Additionally, while the model incorporates feature importance and SHAP interpretability, the current feature set was selected from operational and compositional variables without incorporating time-series or temporal dynamics. Transient effects, such as choke changes, pressure build-ups, or shut-ins, are not explicitly modeled, but they can strongly influence gas rate behavior in the short term. Including such temporal features or moving to recurrent architectures could enhance responsiveness in future work.

Lastly, real-time deployment considerations were beyond the scope of this study. Integration into SCADA systems, sensor latency handling, and model retraining workflows remain areas for future development. However, the modular architecture and computational efficiency of the current model make it well-suited for edge deployment or digital twin platforms. Lightweight API interfaces or deployment on digital twins in real-time systems can help operationalize the model. Real-time retraining loops with SCADA updates can enhance responsiveness.

In summary, while the PIML framework offers notable advantages in terms of accuracy, interpretability, and physics-consistency, it represents an initial step toward a more generalizable and operationally robust flow rate estimation pipeline. Future work should expand scope, address uncertainty modeling at a deeper level, and explore full-cycle deployment scenarios. Feedback loop can be built by coupling the model with downstream optimization engines to close the loop between prediction and control, allowing the system to self-adjust to field interventions or anomalous conditions.

Summary and Conclusions

This paper presents a comprehensive gas rate prediction methodology that fuses empirical physics with modern ensemble machine learning under a physics-informed framework. A recalibrated Modified

Gilbert equation was embedded into a custom loss function, guiding model training to honor physical laws while leveraging the flexibility of data-driven learning. Feature engineering strategies incorporated domain knowledge into the input space, while model architecture (stacked XGBoost–MLP–Ridge) enabled capturing complex nonlinear interactions.

The hybrid PIML model outperformed traditional choke-based equations and displayed performance comparable to standalone ML models, with an R^2 of 0.973 and low residual bias across operational flow ranges. Uncertainty quantification using bootstrapped intervals and quantile regression added interpretability and trustworthiness, while SHAP and feature importance analyses offered transparency into variable contributions.

The study demonstrates that integrating physical insight into ML pipelines leads to scalable, interpretable, and deployable tools for production forecasting. The modular PIML architecture allows flexible adaptation to varying flow regimes and fluid compositions. Most importantly, the physics-informed constraint prevents overfitting, a frequent issue in data-limited oilfield settings.

In conclusion, this work provides a validated template for hybrid gas flow prediction and lays the groundwork for next-generation virtual flow metering solutions. Future work may extend this approach to broader datasets, incorporate geomechanical constraints, or explore integration into reservoir simulators or digital twin platforms for full-loop field optimization.

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